



Project Development

– SpheroidMetrics

Progress update - 17/11/2025

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Data collection and Annotation

Batch 2 data to be collected by 21-11-2025

- Repeat the same workflow for the second batch of annotated data from Dalia.
 - Perform **data augmentation** (horizontal & vertical flips, 180° rotation) to expand the training dataset.

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Poster

- Finding the suitable tools -
 - Key notes for poster sections Assemble, Lucidchart for creating the flowchart.
- Define the Sections for Poster
 - Problem statement, Objective, Methodology flow chart, project roadmap, (expected) results.
- Prepare the Section contents
- Create methodology flowchart - In progress
- Draft the design and template
- Final alignment and finishing ----- Due on: 21-11-2025(submit by 23/11)

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Implementation & Training

Data preprocessing

- Locate images in images folder and masks in labels folder
- Match each image with its mask by filename
- Split dataset into training and validation based on _t suffix
- Resize images and masks to fixed size (squish if needed)
- Normalize image pixels using ImageNet mean and std
- including Gaussian blur (kernel size 25, $\sigma \sim U[0.001, 2.0]$), random affine transforms (translations $\leq 25\%$, rotations $\leq 180^\circ$, scaling 75–125%, shear ≤ 17.5)
- Load images and masks in batches for training.

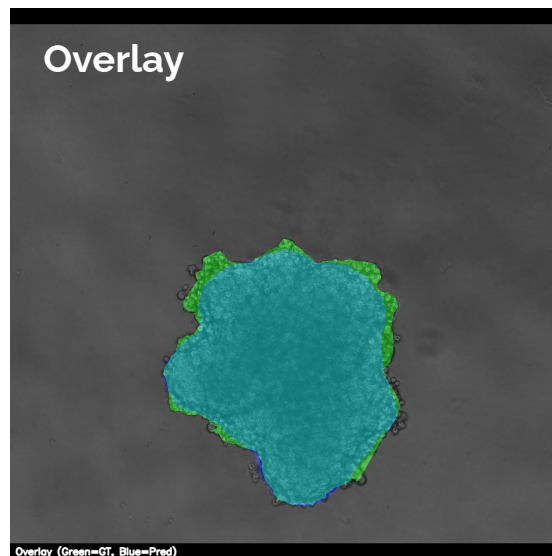
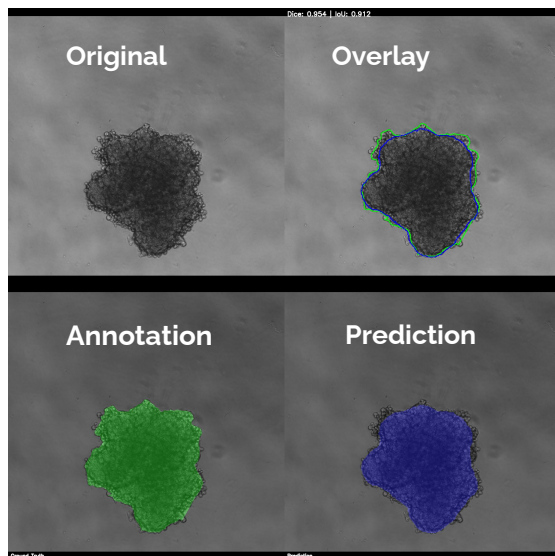
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Implementation & Training

- **U-net**
 - setup the environment,
 - Debug and refactoring the pipeline
 - Train the model on existing data.
- **YOLO**
 - setup the environment,
 - create pipeline
 - Training in progress

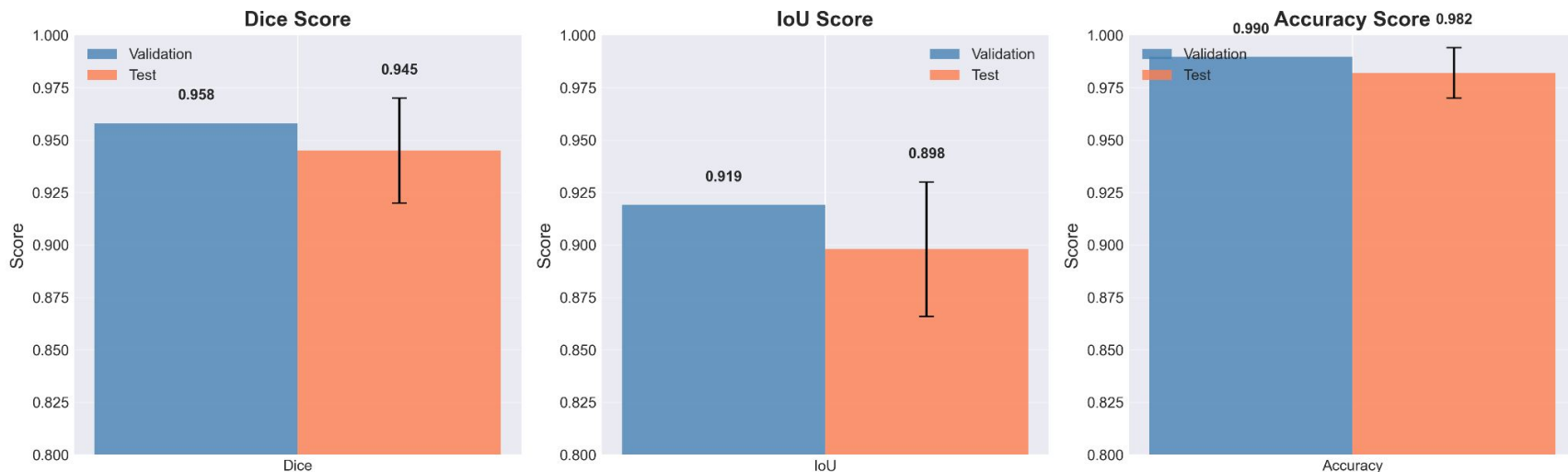
Implementation & Training

Unet - Outcome



Model evaluation metrics

Validation vs Test Performance - unet_resnet34_spheroid



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Implementation - 24-11-2025

- Manually check data for inconsistencies or repetitions
- Complete training for YOLO
- If we get the more data by 21-11-2025
 - Repeat data preprocessing and splitting
 - Retrain the models with the complete dataset
- Otherwise
 - Fine tune the current models
 - Complete the training with current set of data

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Morphological Metrics

- Research and decide on geometrical metrics to use for comparison
- Write script for calculating the decided upon geometrical metrics

Project timeline (High level plan)

